Reference-Free Reliability Scores Cannot See Corrupted Evidence: A Controlled Test for Numerical Retrieval-Augmented QA in Finance

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Abstract

Reference-free scores for retrieval-augmented generation judge a response against the evidence that was retrieved, without the gold answer. Their structural limit is plain: if the retrieved evidence is wrong, a faithful answer is still wrong, and a score computed against that evidence cannot tell. We test this directly for numerical financial question answering. We replace a controlled share of the numbers in the retrieved evidence with wrong values and ask which measures notice, using FinQA and TAT-QA and three generators from two model families (Qwen2.5 3B and 7B, Llama-3.1 8B). Accuracy falls toward zero. Numeric-overlap support, entailment support, lexical groundedness, a hallucination flag and verbalized confidence move by at most a few points and barely separate a fully corrupted case from a clean one (AUROC at most 0.58 in any run; pooled numeric-overlap support 0.53 on FinQA and 0.53 on TAT-QA). Support computed against the *uncorrupted* evidence does (AUROC 0.92 and 0.89), but needs the reference these scores are meant to replace. As detectors of wrong answers and of real retrieval misses the same measures are weak and inconsistent: near chance when pooled on FinQA, somewhat better on TAT-QA, and with signs that differ between models. Along the way we correct a scoring flaw in an earlier version of this work, show that an evidence-noise manipulation we first used did not degrade the model and therefore tested nothing, and show that composite-score rankings built from five-point correlations or weight sweeps are unstable. Code, per-response outputs and a pipeline that reproduces every result on free CPU runners are public.

1 Introduction

Retrieval-augmented generation (RAG) is the standard way to ground a language model's answers about a document, and in finance the answer is usually a number: a ratio, a change, a total. A wrong number that is stated with confidence is costly, so a practical question is whether the system can tell when its own answer is not to be trusted, without the gold answer. A family of *reference-free* scores tries to do this. They check whether the numbers in the response appear in the retrieved evidence, whether the evidence entails the response, or whether the model's stated confidence matches its accuracy (Es et al., 2024; Saad-Falcon et al., 2024; Min et al., 2023). Several such signals are often folded into one composite, so that a single figure summarizes reliability.

This paper examines one such composite for numerical RAG, the Overall Numerical Reliability Score (ONRS), which weights accuracy, numeric evidence support, a hallucination rate and the calibration of verbalized confidence. We ask a narrow question: *when the retrieved evidence is wrong, does a reference-free measure notice?* The question has a structural answer that we state first and then test. A measure that compares the response with the evidence *that was supplied* can only report whether the response is faithful to that evidence. If the evidence itself has been corrupted, a faithful response is still wrong, and the measure has no way to see it.

The test needs a manipulation that actually degrades the model. Our first attempt did not have one. We began by adding unrelated sentences to the evidence and tracking the composite against the amount added, and we initially read the flat result as a failure of the composite. A manipulation check at larger scale showed that the added sentences did not hurt the model at all, so that experiment could not test anything (Section 5). We therefore replace a controlled share of the numbers in the retrieved evidence with wrong ones. This does hurt: accuracy falls to near zero at full corruption. It also lets us ask which measures move.

Contributions.

- A controlled corrupted-evidence test of reference-free reliability measures on two numerical financial QA datasets (FinQA and TAT-QA) with three generators from two model families (Qwen2.5 3B and 7B, Llama-3.1 8B), and smaller models in the secondary analyses. Measures that use only the response and the supplied evidence (numeric-overlap support, entailment support, lexical groundedness, a hallucination flag, verbalized confidence) barely distinguish a fully corrupted case from a clean one (AUROC at most 0.58 in any run). A score that compares the response with the *uncorrupted* evidence can (AUROC 0.85 to 0.93), but it needs the reference we are trying to avoid.
- As detectors of wrong answers on clean evidence, and of real retrieval misses (the retriever fails to surface the numbers the answer needs), the same measures are weak and inconsistent: near chance when pooled on FinQA, modestly informative on TAT-QA, and with effects that change sign between smaller and larger models.
- A correction to a published scoring flaw in this line of work (the model's own confidence statement was counted as unsupported evidence), a demonstration that a five-point correlation and a weight sweep are too unstable to rank composites, and a recommendation that any evidence-quality manipulation be validated by its effect on the model before it is used to judge a metric.
- All code, per-response outputs and the free CPU pipeline that produced them are public.[1]

2 Related Work

Reference-free evaluation of RAG. RAGAs and ARES score faithfulness, answer relevance and context relevance, using a language model as judge or a fine-tuned classifier (Es et al., 2024; Saad-Falcon et al., 2024). Their faithfulness component checks the answer against the context that was retrieved. That is the property whose limit we examine: it is silent about whether the context is right. Atomic-claim scoring (Min et al., 2023) and sampling-consistency checks (Manakul et al., 2023) target hallucination in open-ended text. Natural-language-inference models have been used to score factual consistency between a source and a generated text (Maynez et al., 2020; Laban et al., 2022; Zha et al., 2023); we use an NLI cross-encoder as one of our measures.

Confidence and calibration. Expected calibration error (Guo et al., 2017) summarizes the gap between confidence and accuracy. Language models can verbalize a confidence (Kadavath et al., 2022; Xiong et al., 2024), but the figure is often poorly informative; we find the same for the small models studied here.

Financial numerical QA. FinQA (Chen et al., 2021) asks multi-step arithmetic questions over earnings-report pages, with annotated gold evidence. TAT-QA (Zhu et al., 2021) pairs tables with paragraphs and includes arithmetic, span and counting questions. Both are usually evaluated with gold evidence supplied; we use them in a retrieval setting (Lewis et al., 2020) where retrieval errors occur naturally.

3 Setup

3.1 System

Each report (paragraphs and table rows) is split into chunks, one per sentence or table row, embedded with all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), and the eight chunks most similar to the question are retrieved by exact cosine search. A generator receives a fixed system prompt, two worked examples and the retrieved evidence, and must write its reasoning, a final answer and a self-reported confidence. A rule-based verifier may trigger one retry when the response's own arithmetic does not check out. Decoding is greedy unless stated. Generators are Qwen2.5-1.5B, -3B and -7B-Instruct (Qwen Team, 2024), SmolLM2-1.7B-Instruct (Allal et al., 2025) and Llama-3.1-8B-Instruct. The corrupted-evidence test uses the 3B, 7B and Llama models; the 1.5B and SmolLM2 runs enter only the error-detection and retrieval-miss analyses. The 1.5B, 3B and SmolLM2 models run in float32 on CPU; the 7B and 8B models run as 4-bit GGUF builds (Q4_K_M) through llama.cpp (Gerganov and contributors, 2023), which also resembles the 4-bit setting of earlier runs of this project.

3.2 Data

FinQA: numeric questions from the dev and test splits (the pipeline treats yes/no questions as non-numeric and excludes them from accuracy). **TAT-QA**: the arithmetic questions with a numeric an-

[1] https://github.com/aaronsunny007/Financial-NRF

swer, converted to the same item format; TAT-QA's own test labels are hidden, so its train file serves as our *dev* split and its dev file as our *test* split. Percentage answers are compared as percentages. Accuracy uses a 2% relative (or 0.01 absolute) tolerance on the final answer.

3.3 Measures under test

Numeric-overlap support (NESS) the share of the numbers in the response's reasoning and answer that occur in the supplied evidence. A response is flagged as hallucinated when this falls below 0.5. The model's own confidence statement is excluded from the count (Section 5).

Entailment support each number-bearing, non-arithmetic sentence of the reasoning is the hypothesis and the evidence, in windows of at most 280 words, is the premise. We take the maximum entailment probability over windows from the cross-encoder nli-deberta-v3-small (He et al., 2021) and report the share of claims with probability at least 0.5.

Lexical groundedness the share of the reasoning's content words (four or more letters) found in the evidence. A cheap stand-in for overlap-style faithfulness scoring.

Hallucination flag, self-reported confidence as defined above.

Two further quantities are *not* reference-free and serve as reference points: accuracy (needs the gold answer) and support *against the clean evidence* (needs the uncorrupted evidence). ONRS $=$ $0.30\,\text{acc} + 0.30\,\text{NESS} + 0.20\,(1 -$ hallucination rate$) + 0.20\,(1 - \text{ECE})$, with the ECE computed on the verbalized confidence.

3.4 Manipulations

Corrupted evidence. After retrieval, a fraction $p \in \{0, 25, 50, 75, 100\}\%$ of the numeric tokens in the evidence is replaced by a wrong value (the original multiplied by 0.5, 0.8, 1.25, 1.6, 2 or 3, chosen deterministically per token), preserving currency signs, percent signs, thousands separators and decimals; four-digit years are left alone. Every other part of the pipeline is unchanged. **Retrieval miss.** A natural condition: an item is a *miss* when the retrieved evidence lacks at least one number of the dataset's gold evidence (all FinQA gold facts; for TAT-QA the annotated paragraphs and, if the answer uses the table, the whole table).

3.5 Statistics

Every interval is a 95% percentile bootstrap over items (2000 resamples; the rows belonging to one item stay together). The detection tasks are scored with AUROC, the probability that a randomly chosen positive case scores higher than a randomly chosen negative one (0.5 is chance).

4 Results

4.1 Corrupted evidence

Figure 1 shows how each quantity changes as more of the numbers in the retrieved evidence are replaced. Accuracy falls steeply: averaged over runs, from 0.36 to 0.06 on FinQA and from 0.46 to 0.02 on TAT-QA. Numeric-overlap support computed against the evidence the model was shown barely changes (FinQA 0.77 to 0.76, TAT-QA 0.77 to 0.76), as does entailment support and the hallucination flag. The reason is mechanical: the model copies the corrupted figures it was shown into its reasoning, and those figures are in the evidence. Support computed against the *clean* evidence falls from 0.77 to 0.39 on FinQA and from 0.77 to 0.45 on TAT-QA, because the copied figures are not in the clean evidence.

Table 1 gives the paired change per item from 0% to 100% corruption. Accuracy falls by 0.19 to 0.58 depending on run, and support against the clean evidence by 0.29 to 0.41. The gold-free measures move by at most a few points. For Qwen2.5-7B on FinQA, numeric-overlap and entailment support decline slightly but reliably (about -0.03 and -0.09), an order of magnitude less than accuracy; in every other run, including Llama-3.1-8B, their changes are indistinguishable from zero or of the same small size.

The question is whether a measure can *tell* the two situations apart, not only whether its average moves. Table 2 reports the AUROC for separating a response generated from fully corrupted evidence from one generated from the clean evidence of the same item. Every gold-free measure is close to chance: pooled over runs, numeric-overlap support gives 0.53 [0.50, 0.56] on FinQA and 0.53 [0.50, 0.56] on TAT-QA, entailment support 0.53 and 0.50, and the hallucination flag 0.51 and 0.50. No single run exceeds 0.58 for any gold-free measure. Accuracy, which needs the gold answer, reaches 0.65 and 0.72 (it is limited because many clean responses are already wrong), and support against the clean evidence reaches 0.92 and 0.89. The in-

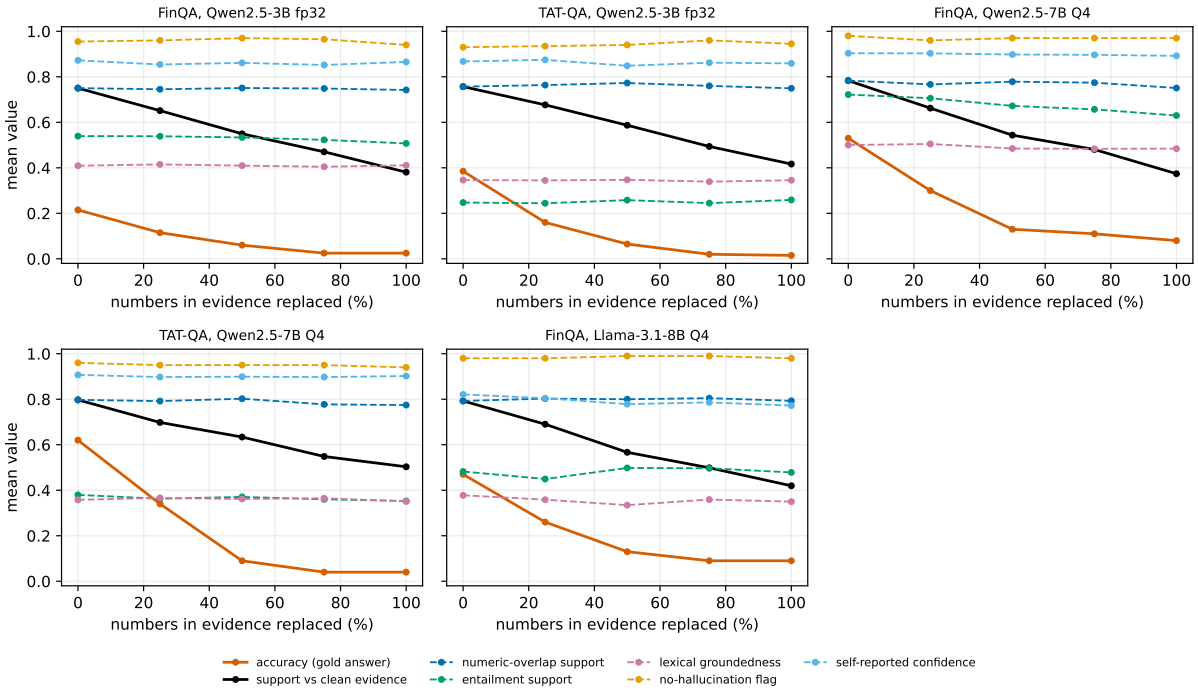


Figure 1: Mean value of each measure against the share of numbers in the retrieved evidence that were replaced by wrong ones (dev and test runs averaged within a dataset and model; 100 items per run). Solid lines need the gold answer or the clean evidence; dashed lines use only the response and the evidence shown to the model.

Run	n	accuracy†	numeric-overlap support	entailment support	hallucination flag	support vs clean‡
FinQA / Qwen2.5-3B fp32 dev	100	-0.19 [-0.27, -0.12]	+0.00 [-0.02, +0.03]	+0.02 [-0.04, +0.07]	+0.01 [-0.04, +0.06]	-0.36 [-0.40, -0.33]
FinQA / Qwen2.5-3B fp32 test	100	-0.19 [-0.27, -0.11]	-0.02 [-0.06, +0.02]	-0.07 [-0.14, -0.00]	+0.02 [-0.03, +0.07]	-0.38 [-0.42, -0.33]
TAT-QA / Qwen2.5-3B fp32 dev	100	-0.37 [-0.46, -0.28]	+0.00 [-0.03, +0.03]	+0.04 [-0.02, +0.10]	-0.02 [-0.07, +0.03]	-0.35 [-0.39, -0.31]
TAT-QA / Qwen2.5-3B fp32 test	100	-0.37 [-0.46, -0.28]	-0.02 [-0.04, +0.01]	-0.02 [-0.08, +0.03]	-0.01 [-0.05, +0.02]	-0.33 [-0.36, -0.30]
FinQA / Qwen2.5-7B Q4 test	100	-0.45 [-0.55, -0.35]	-0.03 [-0.06, -0.01]	-0.09 [-0.16, -0.02]	+0.01 [-0.02, +0.05]	-0.41 [-0.44, -0.37]
TAT-QA / Qwen2.5-7B Q4 test	100	-0.58 [-0.68, -0.47]	-0.02 [-0.05, -0.00]	-0.03 [-0.09, +0.03]	+0.02 [-0.02, +0.06]	-0.29 [-0.33, -0.25]
FinQA / Llama-3.1-8B Q4 test	100	-0.38 [-0.48, -0.28]	+0.00 [-0.03, +0.03]	+0.00 [-0.07, +0.07]	+0.00 [-0.03, +0.03]	-0.37 [-0.41, -0.34]

Table 1: Change from 0% to 100% corruption, paired within item, with 95% bootstrap intervals over items. † needs the gold answer. ‡ needs the clean evidence.

formation that distinguishes the two cases is in the pair (response, clean evidence), which a deployed system does not have.

The composite behaves accordingly. Its correlation with the corruption level over the five levels is between -0.90 and -0.97 in every run, with intervals that exclude zero, and it comes entirely from the accuracy term and from the calibration term, which is a function of accuracy. Replacing numeric-overlap support by support against the clean evidence in the composite moves its correlation with the corruption level only from the range above to between -0.94 and -0.99, because the composite is already dominated by accuracy. So the composite does fall with corrupted evidence, but only through the terms that need the gold answer.

4.2 Detecting wrong answers on clean evidence

On clean evidence the gold-free measures should be useful if they carry information about errors. Table 3 shows they carry little. Pooled over runs, AUROC for separating correct from incorrect answers is 0.47 for numeric-overlap support, 0.55 for entailment support, 0.53 for lexical groundedness, 0.51 for the hallucination flag and 0.54 for verbalized confidence on FinQA (TAT-QA: 0.61, 0.58, 0.56, 0.54, 0.61). Individual runs deviate from 0.5 in both directions. Numeric-overlap support is *below* 0.5 for the 7B and 8B models (0.30 and 0.35 for Qwen2.5-7B on FinQA and TAT-QA, 0.27 for Llama-3.1-8B on FinQA) and above it for the 3B model on TAT-QA (0.71 on dev, 0.74 on test): correct answers contain a computed result that is, by construction, not in the evidence, so how

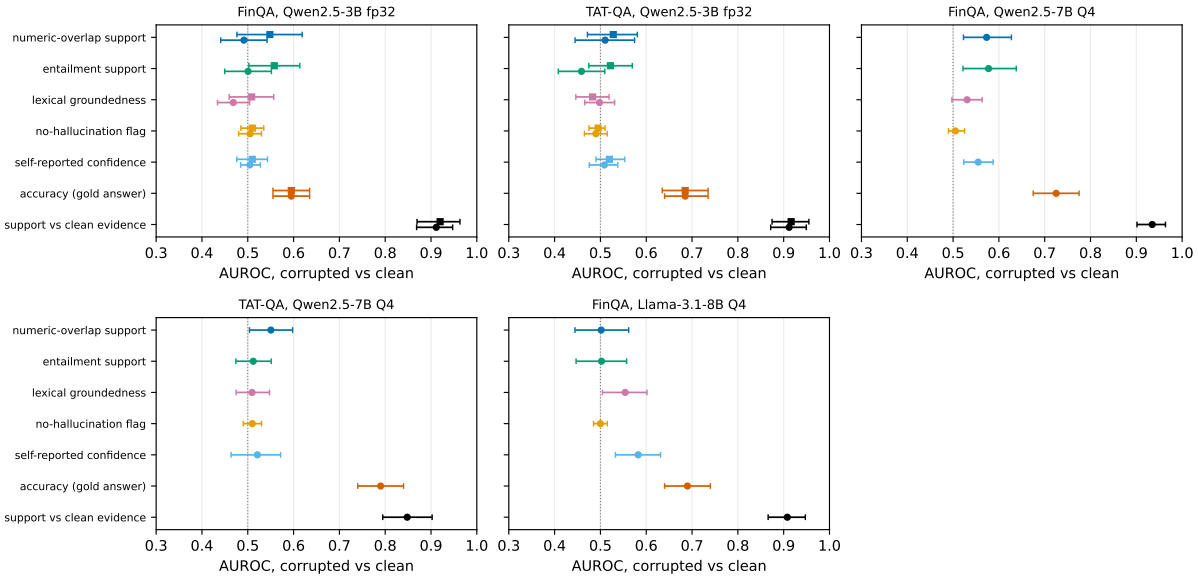


Figure 2: AUROC for telling a response produced from 100%-corrupted evidence from one produced from clean evidence (95% bootstrap intervals over items; one marker per run). The dotted line is chance.

Run	n	numeric-overlap support	entailment support	lexical groundedness	no-hallucination flag	self-reported confidence	accuracy†	support vs clean‡
FinQA / Qwen2.5-3B fp32 dev	100	0.49 [0.44, 0.54]	0.50 [0.45, 0.55]	0.47 [0.43, 0.50]	0.51 [0.48, 0.53]	0.50 [0.48, 0.53]	0.59 [0.56, 0.64]	0.91 [0.87, 0.95]
FinQA / Qwen2.5-3B fp32 test	100	0.55 [0.48, 0.62]	0.56 [0.50, 0.61]	0.51 [0.46, 0.56]	0.51 [0.48, 0.54]	0.51 [0.48, 0.54]	0.59 [0.56, 0.64]	0.92 [0.87, 0.96]
TAT-QA / Qwen2.5-3B fp32 dev	100	0.51 [0.44, 0.57]	0.46 [0.41, 0.51]	0.50 [0.47, 0.53]	0.49 [0.47, 0.52]	0.51 [0.48, 0.54]	0.69 [0.64, 0.73]	0.91 [0.87, 0.95]
TAT-QA / Qwen2.5-3B fp32 test	100	0.53 [0.47, 0.58]	0.52 [0.47, 0.57]	0.48 [0.45, 0.52]	0.49 [0.47, 0.51]	0.52 [0.49, 0.55]	0.69 [0.64, 0.73]	0.92 [0.87, 0.96]
FinQA / Qwen2.5-7B Q4 test	100	0.57 [0.52, 0.63]	0.58 [0.52, 0.64]	0.53 [0.50, 0.56]	0.51 [0.49, 0.53]	0.55 [0.52, 0.59]	0.72 [0.68, 0.78]	0.93 [0.90, 0.96]
TAT-QA / Qwen2.5-7B Q4 test	100	0.55 [0.50, 0.60]	0.51 [0.47, 0.55]	0.51 [0.47, 0.55]	0.51 [0.49, 0.53]	0.52 [0.46, 0.57]	0.79 [0.74, 0.84]	0.85 [0.79, 0.90]
FinQA / Llama-3.1-8B Q4 test	100	0.50 [0.44, 0.56]	0.50 [0.45, 0.56]	0.55 [0.50, 0.60]	0.50 [0.48, 0.52]	0.58 [0.53, 0.63]	0.69 [0.64, 0.74]	0.91 [0.87, 0.95]
FinQA pooled	400	0.53 [0.50, 0.56]	0.53 [0.51, 0.56]	0.52 [0.50, 0.54]	0.51 [0.49, 0.52]	0.54 [0.52, 0.55]	0.65 [0.63, 0.68]	0.92 [0.90, 0.94]
TAT-QA pooled	300	0.53 [0.50, 0.56]	0.50 [0.47, 0.52]	0.50 [0.48, 0.52]	0.50 [0.49, 0.51]	0.52 [0.49, 0.54]	0.72 [0.69, 0.75]	0.89 [0.86, 0.92]

Table 2: AUROC (95% interval) for separating 100%-corrupted from clean evidence. † needs the gold answer. ‡ needs the clean evidence. Pooled rows combine the runs above them.

much weight such a score carries depends on how often the model computes versus copies. Verbalized confidence is nearly constant for the smaller models (most responses say 0.9) and uninformative there, but for the larger models it carries some signal (AUROC 0.73 for Qwen2.5-7B on TAT-QA, 0.65 for Llama-3.1-8B on FinQA), and entailment support reaches 0.70 for Llama-3.1-8B. On TAT-QA the pooled AUROCs for entailment support, lexical groundedness and confidence are 0.58, 0.56 and 0.61, modest but above chance; on FinQA all pooled values stay between 0.47 and 0.55. The lexical baseline performs no better than the entailment or overlap scores, so the weakness is not particular to the numeric-overlap design.

4.3 Real retrieval misses

Corruption is synthetic. A natural failure of the same kind occurs when the retriever does not surface the numbers the answer needs. Table 4 compares items by whether the retrieved evidence contains every number of the gold evidence. The miss rate is 16% on FinQA and 22% on TAT-QA. Accuracy is much lower on misses than on hits. Whether a gold-free measure notices is inconsistent. Pooled over FinQA runs, numeric-overlap support gives AUROC 0.49, entailment 0.57 and lexical groundedness 0.48. On TAT-QA the pooled values are 0.57, 0.61 and 0.54, and verbalized confidence reaches 0.67, driven by the larger models (0.83 for Qwen2.5-7B on TAT-QA, 0.79 for Llama-3.1-8B on FinQA). Numeric-overlap support detects misses for the 3B model on TAT-QA (0.69), plausibly because with the needed figures absent the model substitutes numbers that are then not in the evidence, but the effect reverses for the 7B and 8B models (0.30 and 0.34 for Qwen2.5-7B, 0.38 for Llama-3.1-8B). This is a different failure from corrupted evidence in one respect: *missing* evidence can leave some trace in the response or in the model's own confidence, whereas *wrong* evidence that the model copies leaves almost none. The traces are model-dependent, and these associations are confounded with difficulty (a question

Run	n	Acc.	numeric-overlap support	entailment support	lexical groundedness	no-hallucination flag	self-reported confidence
FinQA / Qwen2.5-3B fp32 dev	100	0.22	0.60 [0.47, 0.73]	0.49 [0.35, 0.62]	0.52 [0.39, 0.65]	0.53 [0.51, 0.56]	0.53 [0.49, 0.59]
FinQA / Qwen2.5-3B fp32 test	97	0.22	0.47 [0.33, 0.61]	0.57 [0.43, 0.71]	0.62 [0.47, 0.75]	0.50 [0.43, 0.54]	0.50 [0.47, 0.53]
TAT-QA / Qwen2.5-3B fp32 dev	100	0.38	0.71 [0.60, 0.81]	0.55 [0.44, 0.67]	0.51 [0.39, 0.63]	0.55 [0.50, 0.60]	0.53 [0.49, 0.58]
TAT-QA / Qwen2.5-3B fp32 test	100	0.39	0.74 [0.64, 0.83]	0.52 [0.42, 0.63]	0.54 [0.43, 0.65]	0.54 [0.51, 0.58]	0.53 [0.48, 0.58]
FinQA / Qwen2.5-7B Q4 test	97	0.55	0.30 [0.20, 0.41]	0.55 [0.44, 0.65]	0.49 [0.37, 0.60]	0.48 [0.45, 0.50]	0.59 [0.52, 0.67]
TAT-QA / Qwen2.5-7B Q4 test	100	0.62	0.35 [0.23, 0.48]	0.61 [0.50, 0.71]	0.63 [0.51, 0.74]	0.51 [0.47, 0.56]	0.73 [0.63, 0.82]
FinQA / Qwen2.5-3B fp32 dev	149	0.26	0.64 [0.54, 0.74]	0.53 [0.42, 0.63]	0.48 [0.38, 0.58]	0.53 [0.51, 0.56]	0.53 [0.49, 0.57]
FinQA / Qwen2.5-3B fp32 test	145	0.21	0.52 [0.40, 0.64]	0.55 [0.43, 0.67]	0.57 [0.44, 0.69]	0.51 [0.47, 0.54]	0.53 [0.49, 0.58]
FinQA / Qwen2.5-1.5B fp32 dev	100	0.40	0.44 [0.32, 0.56]	0.51 [0.41, 0.63]	0.46 [0.34, 0.58]	0.52 [0.47, 0.56]	0.53 [0.44, 0.62]
FinQA / Qwen2.5-1.5B fp32 test	97	0.34	0.48 [0.36, 0.60]	0.50 [0.38, 0.61]	0.48 [0.36, 0.61]	0.54 [0.51, 0.58]	0.55 [0.45, 0.65]
FinQA / SmolLM2-1.7B fp32 dev	100	0.16	0.32 [0.18, 0.47]	0.42 [0.27, 0.59]	0.36 [0.23, 0.51]	0.47 [0.37, 0.55]	0.50 [0.50, 0.50]
FinQA / Llama-3.1-8B Q4 test	97	0.48	0.27 [0.18, 0.38]	0.70 [0.60, 0.80]	0.55 [0.43, 0.66]	0.48 [0.45, 0.50]	0.65 [0.56, 0.73]
FinQA pooled	982	0.31	0.47 [0.43, 0.51]	0.55 [0.52, 0.59]	0.53 [0.49, 0.57]	0.51 [0.50, 0.52]	0.54 [0.52, 0.56]
TAT-QA pooled	300	0.46	0.61 [0.54, 0.67]	0.58 [0.52, 0.64]	0.56 [0.50, 0.63]	0.54 [0.51, 0.56]	0.61 [0.57, 0.65]

Table 3: AUROC for separating correct from incorrect answers under clean evidence (numeric questions only; 95% bootstrap intervals over items).

whose evidence is missing is also a harder question). Accuracy itself separates hits from misses at least as well (pooled FinQA 0.66, TAT-QA 0.73).

5 Correcting the earlier version

This study began as an evaluation of the ONRS composite on FinQA using 4-bit models on a consumer GPU. Re-running it with more items and a test split exposed three problems, which we report because the earlier numbers are public.

1. **A scoring flaw.** The evidence-support score counted every number in the response, including the one in the model's own "Confidence: 0.9" line, which no evidence can support. Every response carried one extra unsupported number, lowering all support scores and pushing some under the hallucination threshold on that number alone. After removing the confidence statement, the hallucination rates reported earlier (0.08 in the baseline, $0.10 \rightarrow 0.20$ under noise) are 0.00 in every condition.

2. **A manipulation that did not work.** The noise experiment added up to four sentences about unrelated figures. At 150 items per split the paired per-item change in accuracy between 0 and 40% noise was $+0.007$ (95% interval -0.05 to $+0.07$) on dev and -0.027 (-0.08 to $+0.02$) on test. The injected text reached the model but did not hurt it, so a metric could not be faulted for failing to track it. Corrupting the numbers of the evidence (Section 3) is the replacement.

3. **Unstable summary statistics.** The original conclusion that the composite fails to track noise and that no weighting of its four components rescues it came from a correlation over five noise-level means and a sweep of 1,771 weightings. On the same 3B model the correlation is $+0.09$ on dev and -0.64 on test, each with an interval of about ± 0.8, and the share of weightings passing both of the original falsification tests is 0.0% on dev and 41.2% on test. The results do not support a ranking of composites.

A statement from the earlier version survives: evidence-support measures that compare a response with the supplied evidence cannot detect corrupted evidence. It was an argument then; Section 4 makes it an experiment. The full list of corrected claims is in the repository (`ERRATUM.md`).

6 Discussion

What the result does and does not say. The corrupted-evidence test does not show that overlap or entailment scores are useless. They measure faithfulness to the supplied context, and they do that job: a response that invents a number is flagged. The finding is about what that property cannot cover. When the failure is in the evidence, any score defined relative to the evidence reports "faithful" and is wrong about reliability. A composite that adds such a score to accuracy and calibration moves with corruption only through the two terms that depend on the gold answer, and so is no better than accuracy at a task where accuracy is unavailable.

Implications for practice. (i) Detecting bad evidence needs a signal about the evidence: agreement between independent retrievals or sources, checks of retrieved figures against a structured record, or a verifier for the evidence rather than for the response. (ii) Our clean-evidence reference scores well, which shows the information exists; the cost is access to a trusted copy. (iii) Validate an evidence-quality manipulation by its effect on the model before using it to evaluate a metric. The

Run	n	Miss	Acc. hit / miss	numeric-overlap support	entailment support	lexical groundedness	no-hallucination flag	self-reported confidence	accuracy†
TAT-QA / Qwen2.5-3B fp32 dev	100	0.27	0.48 / 0.11	0.64 [0.51, 0.78]	0.52 [0.40, 0.63]	0.50 [0.37, 0.63]	0.57 [0.49, 0.65]	0.57 [0.47, 0.67]	0.68 [0.59, 0.76]
TAT-QA / Qwen2.5-3B fp32 test	100	0.20	0.45 / 0.15	0.69 [0.53, 0.83]	0.63 [0.51, 0.73]	0.51 [0.36, 0.65]	0.53 [0.47, 0.61]	0.58 [0.46, 0.71]	0.65 [0.56, 0.74]
FinQA / Qwen2.5-7B Q4 test	100	0.17	0.63 / 0.06	0.30 [0.16, 0.45]	0.65 [0.51, 0.78]	0.61 [0.45, 0.76]	0.49 [0.47, 0.50]	0.58 [0.43, 0.75]	0.78 [0.70, 0.85]
TAT-QA / Qwen2.5-7B Q4 test	100	0.20	0.76 / 0.05	0.34 [0.18, 0.52]	0.69 [0.59, 0.80]	0.62 [0.48, 0.74]	0.54 [0.48, 0.61]	0.83 [0.70, 0.94]	0.86 [0.78, 0.92]
FinQA / Qwen2.5-3B fp32 dev	150	0.13	0.29 / 0.00	0.53 [0.37, 0.68]	0.60 [0.48, 0.72]	0.48 [0.34, 0.62]	0.53 [0.47, 0.61]	0.61 [0.53, 0.71]	0.65 [0.60, 0.68]
FinQA / Qwen2.5-3B fp32 test	150	0.20	0.23 / 0.07	0.59 [0.46, 0.72]	0.49 [0.38, 0.60]	0.43 [0.31, 0.56]	0.51 [0.47, 0.57]	0.47 [0.38, 0.55]	0.58 [0.52, 0.64]
FinQA / Qwen2.5-1.5B fp32 dev	100	0.12	0.44 / 0.08	0.51 [0.32, 0.69]	0.47 [0.30, 0.65]	0.43 [0.24, 0.63]	0.51 [0.44, 0.61]	0.36 [0.23, 0.51]	0.68 [0.57, 0.76]
FinQA / Qwen2.5-1.5B fp32 test	100	0.17	0.39 / 0.06	0.57 [0.42, 0.72]	0.58 [0.43, 0.73]	0.48 [0.33, 0.63]	0.54 [0.47, 0.63]	0.44 [0.30, 0.57]	0.66 [0.58, 0.73]
FinQA / SmolLM2-1.7B fp32 dev	100	0.12	0.18 / 0.00	0.47 [0.29, 0.66]	0.56 [0.43, 0.70]	0.44 [0.27, 0.61]	0.45 [0.42, 0.48]	0.50 [0.50, 0.50]	0.59 [0.55, 0.63]
FinQA / Llama-3.1-8B Q4 test	100	0.17	0.55 / 0.06	0.38 [0.22, 0.53]	0.69 [0.55, 0.80]	0.50 [0.36, 0.63]	0.49 [0.47, 0.50]	0.79 [0.65, 0.91]	0.75 [0.66, 0.82]
FinQA pooled	800	0.15	0.37 / 0.05	0.49 [0.42, 0.54]	0.57 [0.51, 0.62]	0.48 [0.42, 0.54]	0.51 [0.49, 0.53]	0.53 [0.48, 0.58]	0.66 [0.63, 0.69]
TAT-QA pooled	300	0.22	0.57 / 0.10	0.57 [0.47, 0.66]	0.61 [0.54, 0.68]	0.54 [0.46, 0.61]	0.55 [0.51, 0.59]	0.67 [0.60, 0.74]	0.73 [0.68, 0.78]

Table 4: Items split by whether the retrieved evidence contains every number of the gold evidence (FinQA: annotated gold facts; TAT-QA: operands of the annotated derivation). AUROC: can the measure tell complete from incomplete retrieved evidence? † needs the gold answer.

distractor sentences we first used would have been reported as a metric failure.

7 Conclusion

Reference-free reliability scores for numerical RAG are blind to corrupted evidence by construction, and a controlled test on two datasets and three generators from two model families confirms it: accuracy collapses while every measure that sees only the response and the supplied evidence stays near chance (AUROC at most 0.58). The same measures are weak and model-dependent at flagging wrong answers on clean evidence and real retrieval misses. Practitioners who need reliability estimates should treat such scores as faithfulness checks and add a signal about the evidence itself.

Limitations

The generators are small, open models (1.5B to 8B), run without fine-tuning; larger or instruction-tuned-for-finance models may express confidence differently. Accuracy is low for the smaller models (about 0.2 on FinQA for the 3B model; 0.47 to 0.62 for the 7B and 8B models), which limits the power of the correct-versus-incorrect comparisons; we report intervals instead of claiming equality. Corruption is synthetic and numeric; it isolates the structural point but is not a model of how real filings are wrong. The retrieval-miss analysis defines a miss by the absence of gold numbers in the retrieved text, which can disagree with whether the answer was recoverable; the TAT-QA oracle is document-level because the dataset has no cell-level evidence labels. The entailment measure uses one small NLI model and the lexical baseline is a heuristic; neither is a model-based RAGAs judge, which we could not run for free and do not claim to reproduce. The 7B and 8B models are 4-bit builds run through a different inference engine than the float32 models, so differences between sizes are confounded with precision. We do not report a GPU 4-bit run of the smaller models. Item counts (100 per run) are modest; pooling over runs narrows the intervals but mixes models.

Reproducibility

Code, prompts, per-response outputs for every run, and the GitHub Actions workflow that regenerates them on free CPU runners are at `https://github.com/aaronsunny007/Financial-NRF`. Tables and figures in this paper are produced from the result files by `experiments/paper_analysis.py` and `experiments/paper_figures.py`.

Acknowledgements

The experiments, analysis code and a first draft of this text were produced with the assistance of Claude (Anthropic), working in an agentic coding environment. The author directed the study, reviewed the results and the text, and is responsible for all claims. No external funding supported this work.

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